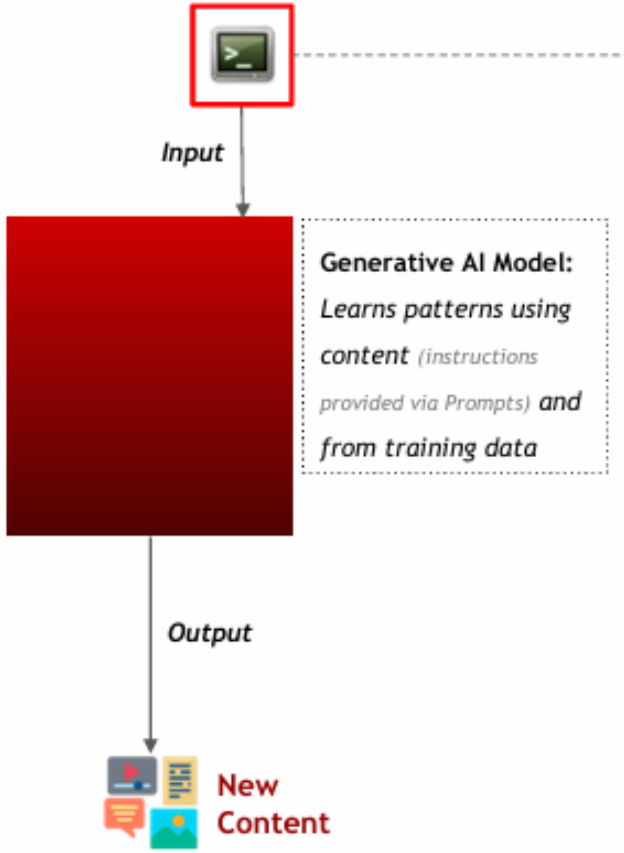
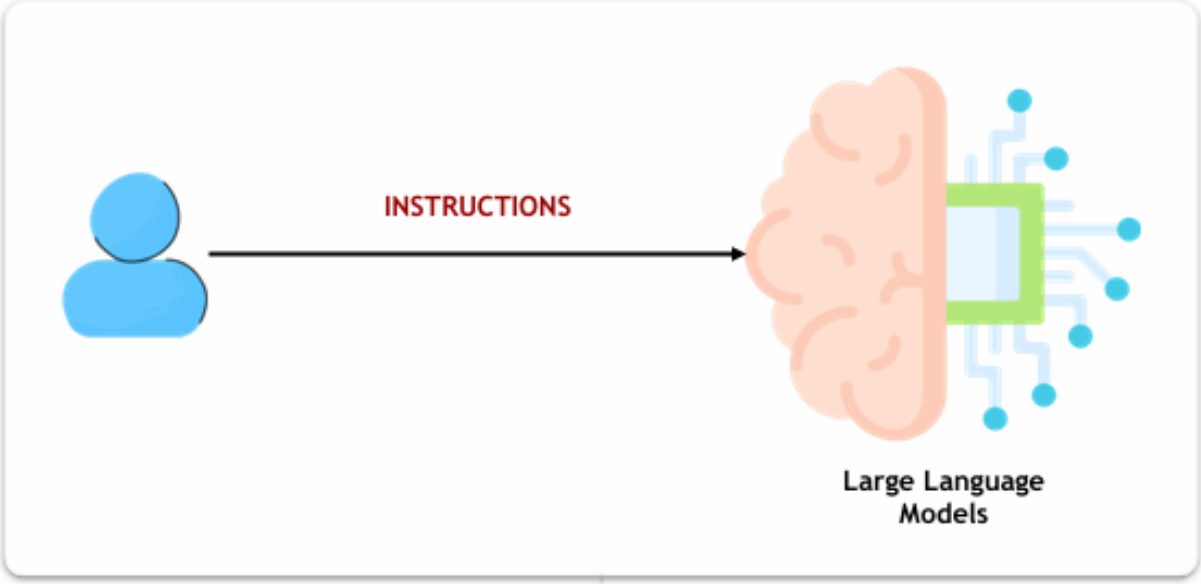
Prompt Engineering

What is a Prompt?



- *A prompt refers to a specific instruction, query, or input given to an AI model to generate a response or perform a task*
- *It serves as cue, guidance, or a starting point for the AI system to produce an output based on its learned knowledge and language comprehension*
- *The form and tone of a prompt may vary depending on the AI application and the desired outcome*
- *Well-defined, context-driven, and unambiguous prompts bring AI output closer to the intended results*

What is Prompt Engineering?



Prompt Engineering is an art of asking the right question to get the best response from an LLM

What is Prompt Engineering?

Developer

As per me,

PROMPT	*Code blocks, individual lines of code, or natural language comments a developer writes to generate a specific suggestion from GitHub Copilot*
PROMPT ENGINEERING	*Providing instructions or comments in the IDE to generate specific coding suggestions*
CONTEXT	*Details that are provided by a developer to specify the desired output from a generative AI coding tool*

Why is Prompt Engineering Important?

- Improve the Accuracy
- Improve the usefulness of AI generated content
- Reduce the risk of generating harmful and biased content

Key elements of a Prompt

Let's look at aspects that make up a good prompt:

Task

Clearly specify the desired action or response from the model

Context

Provide relevant background information to help the model understand the setting or scenario

Exemplar

Offer a sample of the desired output to guide the model

Persona

Define the character or role the model should respond to

Format

Specify the structure or style of the response

More Important

Optional

Key elements of a Prompt

Let's look at aspects that make up a good prompt:

You are a senior software architect with over 15 years of experience in developing secure and scalable web applications. I am a mid-level developer, working in a fast-paced startup environment, and I need to create a REST API for user authentication using Flask. I have experience with Django but am new to Flask and handling large-scale authentication systems.

Your task is to guide me through building a simple but secure authentication API. Please provide clear, step-by-step instructions. The goal is to ensure the API has endpoints for registration, login, and logout, implements token-based authentication, and includes proper error handling.

Structure the response as follows:

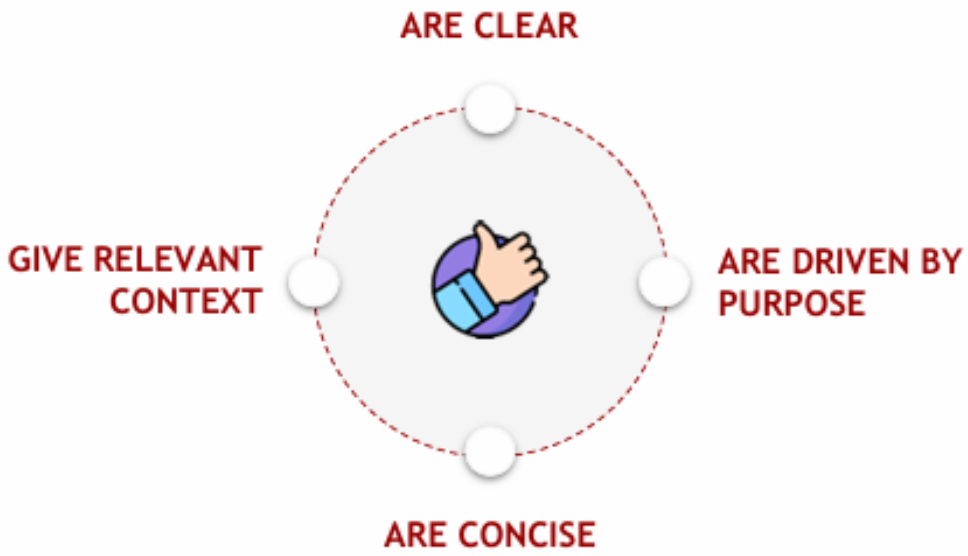
- ***Introduction:*** *A brief overview of user authentication and why Flask is suitable for this task.*
- ***Requirements:*** *List of libraries and tools needed.*
- ***Step-by-Step Implementation:*** *Break down the process with code snippets and explanations. Include comments within the code for clarity.*
- ***Best Practices:*** *Advice on security measures, error handling, and scalability.*
- ***Sample Code Response:*** *Provide an example response for a successful login and an error case.*

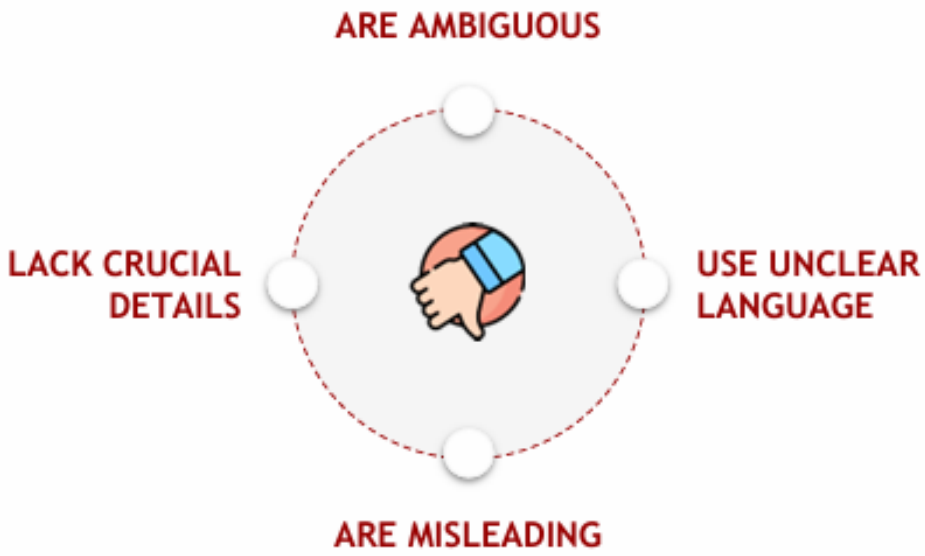
For the 'Step-by-Step Implementation' section, include a sample code snippet. For example, when implementing the login endpoint, show how to handle user input, validate credentials, and return a JSON response with a success message and token

PERSONA CONTEXT TASK FORMAT EXEMPLAR

Good Prompts vs. Bad Prompts

The quality of the prompt significantly influences the output generated by the model. Here are the traits of Good & Bad Prompts:





Tips for designing Prompts

Write Clear Instructions

Help me build an authentication system

Guide me through building a user authentication system in Flask. Include steps for creating endpoints for user registration, login, and logout. Use token-based authentication and include code examples for each step. Make sure to explain how to handle errors and security best practices

Change your prompt as per following to make it clearer:

- *Include details in your query to get more relevant answers*
- *Ask the model to adopt a persona*
- *Use delimiters to clearly indicate distinct parts of the input*
- *Specify the steps required to complete a task*
- *Provide examples*

THANK YOU